

Satya Bhargav Maddipati

sbmaddipati@wisc.edu | GitHub | LinkedIn | Portfolio

M.S. Computer Sciences student at UW-Madison with experience building production AI services, retrieval workflows, and machine learning systems.

EDUCATION

University of Wisconsin-Madison | M.S. in Computer Sciences

Sep 2025 - May 2027

GPA: 4.0/4.0. Coursework: AI, machine learning, computer vision, foundation models, databases, big-data systems, and HCI.

IIT Bombay | B.Tech + M.Tech Dual Degree, Honors

Aug 2018 - Jun 2023

Environmental Science and Engineering. GPA: 8.9/10. Coursework: machine learning, probability, statistics, NLP, and data science.

EXPERIENCE

Tata AIA Life Insurance | Assistant Manager, Data Science

Jul 2023 - Dec 2025

Management Trainee, Jul 2023-Jun 2024; Assistant Manager, Jun 2024-Dec 2025

- Built production GPT-4 and RAG services with FastAPI on Azure, achieving 0.8s p95 latency and 99.5% uptime.
- Developed agentic and retrieval workflows that reduced manual review by 35%, saving an estimated 2,000+ hours annually.
- Created a multi-agent LLM evaluation platform processing 50K+ evaluations monthly with 89% correlation to human scores.
- Improved document-classification accuracy from 78% to 94% by fine-tuning Llama and Qwen with QLoRA and PEFT.

IIT Bombay | Graduate Research Assistant

Apr 2022 - Jun 2023

- Built 3-7 day air-quality forecasts using SARIMA, XGBoost, Random Forests, and meteorological data.
- Applied stochastic KZ filtering and SHAP analysis to explain seasonal and feature-level contributions.

IIT Bombay | Teaching Assistant

Aug 2022 - May 2023

- Led labs, tutorials, and office hours, helping students understand course material and assignments.

Cognibot AI | Machine Learning Intern

Dec 2021 - Feb 2022

- Developed an EfficientDet computer-vision pipeline for tracking people in retail environments across different compute constraints.

SELECTED PROJECTS

Reason-Guessr | PyTorch, CLIP, Computer Vision

Built an interpretable image-geolocation pipeline on a 32K-image dataset, reaching 88.9 km median haversine error.

Ray-Ban AI Life Logger | Python, Whisper, Claude, SQLite

Built an automated pipeline that transcribes captured audio, filters useful signals, and delivers structured hourly summaries.

Grounded Vision-Language Agent | LLaVA-1.6, LoRA/PEFT, Hugging Face

Implemented an Observe-Reason-Act loop for multi-step visual instruction following and evaluated it against text-only baselines.

TECHNICAL SKILLS

Languages: Python, C++, SQL, JavaScript, Java, R, MATLAB

Backend: FastAPI, Flask, REST APIs, PostgreSQL, Redis, SQLite

ML & AI: PyTorch, Scikit-learn, Hugging Face, OpenCV, XGBoost, RAG, LangGraph

Cloud: Docker, Kubernetes, CI/CD, Azure, AWS